	Case Name: Young Cattle Barn (2)	Sector	Construction (institutional building)	
OR-AS Operations Research Applications and Solutions	**OR-AS** Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions	
Date	**2015**	Project Control	Automatic tracking	
File Name	**C2015-01 Young Cattle Barn (2)**		Tracking based on user input	

1. Project description

Project authenticity

The project replaces an old barn with a modern, durable facility featuring essential installations, improved usability, and enhanced animal welfare.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	27
Planned Duration (PD)	131 days*
Budget At Completion (BAC)	612.769,44€
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	57%
Activity Distribution (AD)	73%
Length of Arcs (LA)	0%
Topological Float (TF)	46%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	12	13.4	-0.8
CRI-rho	14	16.9	-0.6
CRI-tau	8	33.3	-1.9

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	51	37.8	0.19
SI	84	36.5	1.5
SSI	14	12.4	-1.6
CRI-r	12	13.0	-1.5
CRI-rho	16	16.7	-0.9
CRI-tau	9	33.5	-1.9

2.3. *Project Control*

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	6.4	-2.4
PV - SPI	10.3	-3.4
PV - SCI	16.6	-4.4
ED - 1	6.4	-2.3
ED - SPI	10.2	-3.5
ED - SCI	13.5	-4.3
ES - 1	5.9	-2.7
ES - SPI(t)	10.6	-4.3
ES - SCI(t)	14.0	-5.2

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	3.9	0
CPI	6.2	-1.8
SPI	7.4	-1.0
SPI(t)	8.3	-1.6
SCI	11.1	-1.8
SCI(t)	12.0	-2.5
0.8 CPI + 0.2 SPI	6.2	0.2
0.8 CPI + 0.2 SPI(t)	6.3	0.3

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 9 tracking periods with a length of approximately 1 per month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-35919.3	-146937.8	-430.8	0.9	0.6	0.5	1.0
std dev	17696.4	115828.5	187.3	0.1	0.2	0.1	0.1
final	-33704.2	0	-631.0	1.0	1	0.6	1.0

2.3.3.2. Time forecasting

PD	131 days	Real Duration	210 days	Late	60.3%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	145.2	2.1	3.4	3.4
PV - SPI	145.8	2.1	3.4	3.4
PV - SCI	146.4	2.1	3.4	3.4
ED - 1	147.2	1.9	3.3	3.3
ED - SPI	148.0	1.9	3.3	3.3
ED - SCI	148.8	1.9	3.2	3.2
ES - 1	149.6	2.1	3.2	3.2
ES - SPI(t)	150.2	2.1	3.2	3.2
ES - SCI(t)	150.8	2.1	3.1	3.1

2.3.3.3. Cost forecasting

BAC	612.769,4€	Real Cost	646.473,6€	Over Budget	5.5%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	648688.7	17696.4	0.0	0.0
CPI	705595.8	50511.6	1.1	-1.1
SPI	926503.7	233539.1	4.8	-4.8
SPI(t)	987269.8	386379.2	5.9	-5.9
SCI	1038923.3	327232.6	6.7	-6.7
SCI(t)	1109968.7	497963.3	8.0	-8.0
0.8 CPI + 0.2 SPI	733831.5	70307.8	1.5	-1.5
0.8 CPI + 0.2 SPI(t)	736927.3	72117.5	1.6	-1.6